

# RW3-0.06

## Interface Control Document

Medium Reaction Wheel - 0.06 Nms

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Hardware Revision 5

|                           |                                |
|---------------------------|--------------------------------|
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| <b>Product:</b>           | RW3-0.06 Medium Reaction Wheel |
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| <b>Prepared By:</b>       | AnyCompany Engineering Team    |
| <b>Approved By:</b>       | Systems Engineering            |

**AnyCompany**

# 1. Document History

| Version | Date       | Author      | Description                                                                  |
|---------|------------|-------------|------------------------------------------------------------------------------|
| 1.0     | 2014-03-15 | J. Kowalski | Initial release for RW3-0.06 production baseline                             |
| 2.0     | 2018-09-22 | M. Chen     | Redesigned protocol from legacy UART to RS-422 differential signaling        |
| 2.1     | 2021-04-10 | S. Patel    | Added redundant RS-485 secondary interface for fault tolerance               |
| 2.2     | 2025-06-10 | R. Tanaka   | Updated fault handling procedures, added event counter, revised CRC to CRC32 |

## 1.1 Applicable Documents

| ID   | Document            | Title                                                                                  |
|------|---------------------|----------------------------------------------------------------------------------------|
| RD-1 | AC-DS-00068         | RW3-0.06 Product Data Sheet                                                            |
| RD-2 | AC-TP-00068         | RW3-0.06 Qualification Test Procedure                                                  |
| RD-3 | AC-UM-00068         | RW3-0.06 User Manual                                                                   |
| RD-4 | EIA/TIA-422-B       | Electrical Characteristics of Balanced Voltage Digital Interface Circuits              |
| RD-5 | EIA/TIA-485-A       | Electrical Characteristics of Generators and Receivers for Balanced Multipoint Systems |
| RD-6 | ECSS-Q-ST-70-02C    | Thermal Vacuum Outgassing Test for Spacecraft Materials                                |
| RD-7 | NASA GSFC-STD-7000A | General Environmental Verification Standard (GEVS)                                     |

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### 3. Scope and Purpose

This Interface Control Document (ICD) defines the mechanical, electrical, environmental, and software interfaces for the AnyCompany RW3-0.06 medium-class reaction wheel, hardware revision 5. It establishes the requirements and constraints that must be satisfied by both the reaction wheel and the host spacecraft to ensure proper integration and operation.

The RW3-0.06 provides 0.06 Nms angular momentum storage with up to 20 mNm of torque output. It is designed for attitude determination and control subsystems (ADCS) on small to medium spacecraft in the 12U to 100 kg class. The wheel features RS-422 primary and redundant RS-485 communication interfaces operating at 460,800 baud, a wide input voltage range of 7.5 to 36 V, and a mass of only 235 g.

With 317 units flown and over 930 cumulative flight years since 2014, the RW3-0.06 is the most-flown model in the AnyCompany reaction wheel product line. This document is applicable to hardware revision 5 and later units.

**Applicable Product Revision:** RW3-0.06 Revision 5 and later

## 4. Mechanical Interface

### 4.1 Physical Specifications

| Parameter                | Value        | Units             | Notes                                        |
|--------------------------|--------------|-------------------|----------------------------------------------|
| Mass                     | 235          | g                 | Including mounting hardware, excluding cable |
| Length                   | 77           | mm                | Along mounting face                          |
| Width                    | 65           | mm                | Perpendicular to mounting face               |
| Height                   | 37.5         | mm                | From mounting face to top of housing         |
| Mounting Bolt Pattern    | 4x M4        | -                 | On 70 mm bolt circle diameter                |
| Mounting Standoff        | 5.0 min      | mm                | Required boss/spacer height                  |
| Center of Mass (X)       | 0.0 +/- 1.0  | mm                | From geometric center                        |
| Center of Mass (Y)       | 0.0 +/- 1.0  | mm                | From geometric center                        |
| Center of Mass (Z)       | 18.5 +/- 1.5 | mm                | From mounting face                           |
| Moment of Inertia (spin) | 9.5e-5       | kg*m <sup>2</sup> | About spin axis                              |

### 4.2 Mounting Interface

The wheel is mounted using four M4 bolts arranged on a 70 mm diameter bolt circle. It is critical that the wheel frame not be warped when mounted, as this can impede proper function of the bearings. A machined boss or spacer having a height of at least 5.0 mm beneath each mounting point is required. Recommended torque: 1.5 Nm with thread-locking compound.

**Remove Before Flight:** Two Kapton tape bearing covers protect the bearings from debris during ground handling. These may be removed prior to flight.

### 4.3 Outline Drawing

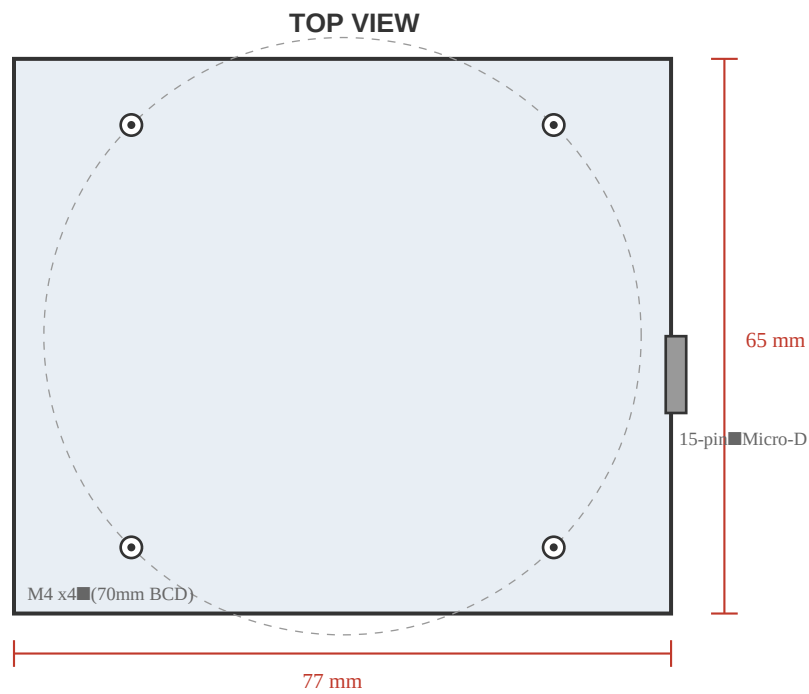


Figure 1: RW3-0.06 Top View Outline Drawing (not to scale)



Figure 2: RW3-0.06 Product Photo

## 5. Environmental Specifications

### 5.1 Thermal

| Parameter                  | Min | Typical | Max | Units | Notes                       |
|----------------------------|-----|---------|-----|-------|-----------------------------|
| Survival Temperature       | -40 | -       | +85 | C     | Non-operating, at interface |
| Operating Temperature      | -30 | +20     | +70 | C     | At baseplate interface      |
| Storage Temperature        | -40 | +22     | +85 | C     | Sealed with desiccant       |
| Thermal Dissipation (idle) | -   | 1.3     | -   | W     | At nominal speed            |
| Thermal Dissipation (peak) | -   | -       | 8.0 | W     | At max torque command       |

Temperature is defined as the temperature of the base plate in the nominal mounting configuration. Operating temperature range has been derated relative to the product qualification range. Contact AnyCompany for extended temperature qualification data.

### 5.2 Pressure and Outgassing

The wheel is qualified for operation in sea-level atmosphere and hard vacuum. It has not been qualified for operation at high-altitude atmospheres and should not be powered during ascent unless additional testing confirms no danger of arcing. All materials meet standard outgassing requirements: TML < 1.0%, CVCM < 0.1% per ECSS-Q-ST-70-02C.

### 5.3 Vibration and Shock

| Frequency Range | Qualification Level     | Acceptance Level        | Duration   |
|-----------------|-------------------------|-------------------------|------------|
| 20 - 50 Hz      | +3 dB/oct ramp          | +3 dB/oct ramp          | 2 min/axis |
| 50 - 800 Hz     | 0.04 g <sup>2</sup> /Hz | 0.02 g <sup>2</sup> /Hz | 2 min/axis |
| 800 - 2000 Hz   | -6 dB/oct rolloff       | -6 dB/oct rolloff       | 2 min/axis |
| Overall         | 14.1 Grms (Qual)        | 10.0 Grms (Accept)      | 3 axes     |

Table 1: Random Vibration Profile per NASA GEVS (GSFC-STD-7000A)

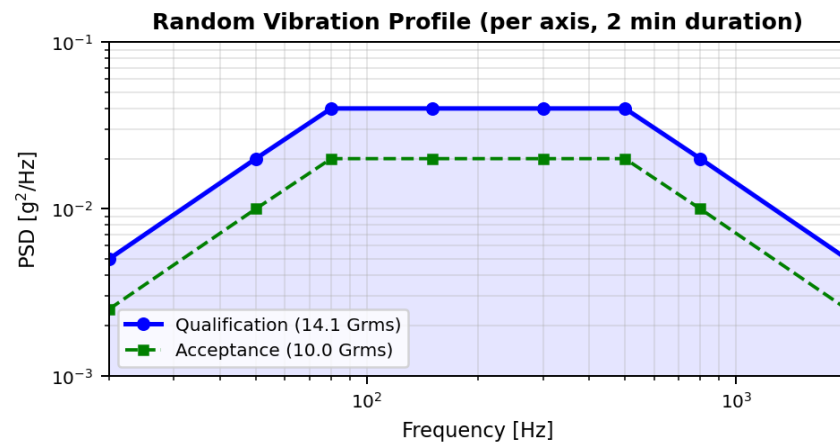


Figure 3: Random Vibration PSD Profile

## 5.4 Radiation

The RW3-0.06 electronics are qualified to a total ionizing dose (TID) of 20 krad at the board level. This applies to the accumulated dose over the mission lifetime. Single-event effects (SEE) testing has been performed; contact AnyCompany for detailed SEE characterization data.

## 6. Electrical Interface

### 6.1 Power Architecture

The RW3-0.06 accepts a wide input voltage range of 7.5 to 36 V (nominal 28 V). The power architecture uses a front-end EMI filter followed by a DC/DC converter that generates regulated internal rails. An H-bridge motor driver controls the three-phase BLDC motor. Regenerative braking returns energy to the bus when decelerating; a bus voltage clamp circuit activates at 38 V to prevent overvoltage in case the bus cannot absorb regenerated energy.

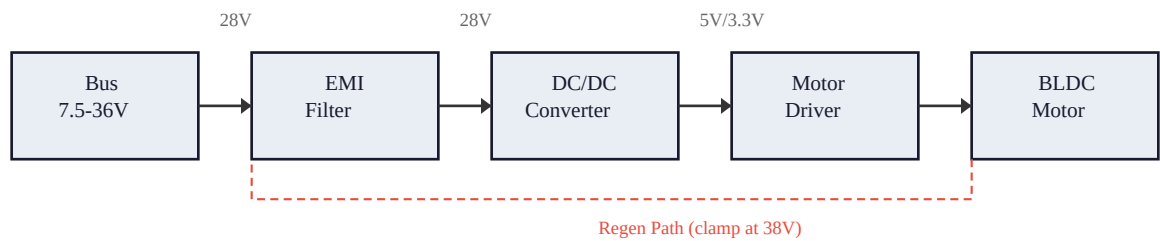


Figure 4: Power Architecture Block Diagram

| Parameter             | Min  | Typical | Max | Units | Notes                         |
|-----------------------|------|---------|-----|-------|-------------------------------|
| Input Voltage         | 7.5  | 28      | 36  | V     | Operating range               |
| Abs Max Voltage       | -0.5 | -       | 40  | V     | Non-destructive               |
| Idle Power            | -    | 1.3     | -   | W     | Steady state at nominal speed |
| Peak Power            | -    | -       | 8.0 | W     | At max torque 20 mNm          |
| Inrush Current        | -    | -       | 2.0 | A     | Duration < 10 ms              |
| Regen Clamp Voltage   | -    | 38      | -   | V     | Bus clamp activation          |
| Front-end Capacitance | -    | 1.0     | -   | uF    | Input filter                  |
| Total Bus Capacitance | -    | 47      | -   | uF    | Total on power rail           |

Table 2: Power Interface Specifications

### 6.2 Connector Pinout

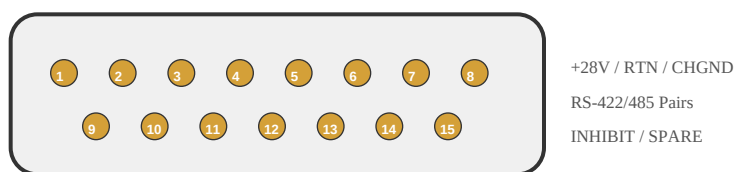
The RW3-0.06 uses a 15-pin Micro-D connector (MIL-DTL-83513 compatible).

| Pin | Signal     | Color | Description                     | Voltage Level |
|-----|------------|-------|---------------------------------|---------------|
| 1   | +28V Power | Red   | Primary power input 7.5-36V     | 7.5 - 36 V    |
| 2   | Power RTN  | Black | Power return (ground reference) | GND           |

| Pin | Signal      | Color  | Description                                    | Voltage Level  |
|-----|-------------|--------|------------------------------------------------|----------------|
| 3   | Chassis GND | Green  | Chassis/EMI ground                             | GND (isolated) |
| 4   | TxA+        | Orange | RS-422 Primary Transmit +                      | Differential   |
| 5   | TxA-        | Brown  | RS-422 Primary Transmit -                      | Differential   |
| 6   | RxA+        | Blue   | RS-422 Primary Receive +                       | Differential   |
| 7   | RxA-        | White  | RS-422 Primary Receive -                       | Differential   |
| 8   | TxB+        | Violet | RS-485 Redundant Transmit +                    | Differential   |
| 9   | TxB-        | Gray   | RS-485 Redundant Transmit -                    | Differential   |
| 10  | RxB+        | Pink   | RS-485 Redundant Receive +                     | Differential   |
| 11  | RxB-        | Tan    | RS-485 Redundant Receive -                     | Differential   |
| 12  | INHIBIT_N   | Yellow | Active-low inhibit (pull low to disable motor) | 0 - 5 V CMOS   |
| 13  | SPARE       | -      | Reserved for future use                        | -              |
| 14  | SPARE       | -      | Reserved for future use                        | -              |
| 15  | SPARE       | -      | Reserved for future use                        | -              |

Table 3: 15-pin Micro-D Connector Pinout

## 6.3 Connector Face View



15-pin Micro-D (Mating Face View)

Figure 5: Connector Face View (mating side)

## 6.4 Interface Electrical Specifications

| Parameter              | Min | Typical | Max    | Units    |
|------------------------|-----|---------|--------|----------|
| Baud Rate              | -   | 460800  | -      | bps      |
| Data Format            | -   | 8N1     | -      | -        |
| Driver Output (diff)   | 2.0 | -       | 5.0    | V        |
| Receiver Input (diff)  | 0.2 | -       | 12.0   | V        |
| Termination Resistance | -   | 120     | -      | Ohm      |
| Max Cable Length       | -   | -       | 15     | m        |
| ESD Protection         | -   | -       | +/- 15 | kV (HBM) |

*Table 4: RS-422/RS-485 Interface Specifications*

## 7. Software Interface

### 7.1 Sign Convention

Positive torque and momentum correspond to counterclockwise (CCW) rotation when viewed from the mounting face (connector side) of the wheel. A positive speed command will accelerate the flywheel in the CCW direction.

### 7.2 Protocol Overview

The RW3-0.06 uses a custom binary packet protocol over RS-422 (primary) or RS-485 (redundant). This is different from the NSP/SLIP protocol used on the smaller wheel models (RW-0.003, RW-0.01, RW-0.03). The binary protocol provides higher throughput and supports CRC-32 error detection.

### 7.3 Packet Structure

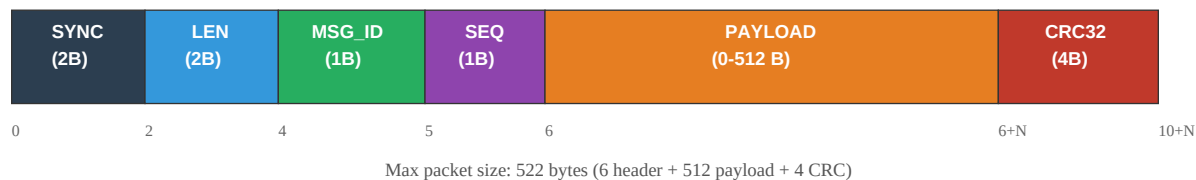


Figure 6: Binary Packet Frame Structure

| Field   | Bytes | Value / Range         | Description                                  |
|---------|-------|-----------------------|----------------------------------------------|
| SYNC    | 2     | 0xEB90                | Synchronization word (fixed)                 |
| LENGTH  | 2     | 0x0000-0x0206         | Payload length in bytes (little-endian)      |
| MSG_ID  | 1     | 0x00-0xFF             | Message identifier                           |
| SEQ     | 1     | 0x00-0xFF             | Sequence counter (wrapping)                  |
| PAYLOAD | 0-512 | Variable              | Message-specific data                        |
| CRC32   | 4     | 0x00000000-0xFFFFFFFF | CRC-32/MPEG-2 over LENGTH+MSG_ID+SEQ+PAYLOAD |

Table 5: Packet Field Definitions

### 7.4 Message Identifiers

| Code | Name       | Dir     | Description                             | Payload (B) |
|------|------------|---------|-----------------------------------------|-------------|
| 0x01 | HEARTBEAT  | CMD/RPL | Keepalive; resets watchdog timer        | 0 / 4       |
| 0x02 | SET_MODE   | CMD     | Set operating mode (see 7.5)            | 1           |
| 0x03 | SET_SPEED  | CMD     | Set wheel speed setpoint (RPM, float32) | 4           |
| 0x04 | SET_TORQUE | CMD     | Set torque command (mNm, float32)       | 4           |

| Code | Name           | Dir     | Description                              | Payload (B) |
|------|----------------|---------|------------------------------------------|-------------|
| 0x10 | TELEM_REQUEST  | CMD     | Request telemetry packet                 | 0           |
| 0x11 | TELEM_RESPONSE | RPL     | Telemetry data (see 7.6)                 | 41          |
| 0x20 | CONFIG_READ    | CMD/RPL | Read configuration parameter by index    | 2 / var     |
| 0x21 | CONFIG_WRITE   | CMD     | Write configuration parameter by index   | var         |
| 0x30 | FW_UPDATE      | CMD     | Firmware update data block               | var         |
| 0xF0 | ERROR          | RPL     | Error report with fault code and context | 8           |

Table 6: Message Identifier Codes

## 7.5 Operating Modes

| Code | Mode        | Description                      | Entry Condition                   |
|------|-------------|----------------------------------|-----------------------------------|
| 0x00 | STANDBY     | Motor disabled, telemetry active | Power-on default; SET_MODE 0x00   |
| 0x01 | SPEED_CTRL  | Closed-loop speed control        | SET_MODE 0x01 + SET_SPEED         |
| 0x02 | TORQUE_CTRL | Open-loop torque command         | SET_MODE 0x02 + SET_TORQUE        |
| 0x03 | SAFE_MODE   | Motor brake, fault recovery      | Automatic on fault; SET_MODE 0x03 |
| 0x04 | FW_UPDATE   | Firmware update mode             | SET_MODE 0x04 + authentication    |

Table 7: Operating Modes

## 7.6 Telemetry Response Packet

The TELEM\_RESPONSE (0x11) packet is transmitted at 10 Hz when telemetry is active.

| Offset | Field            | Type    | Units    | Range               |
|--------|------------------|---------|----------|---------------------|
| 0      | wheel_speed_rpm  | float32 | RPM      | +/- 6000            |
| 4      | motor_current_mA | float32 | mA       | 0 - 500             |
| 8      | bus_voltage_V    | float32 | V        | 0 - 40              |
| 12     | pcb_temp_C       | float32 | C        | -40 to +100         |
| 16     | bearing_temp_C   | float32 | C        | -40 to +100         |
| 20     | fault_register   | uint32  | bitfield | See Section 9       |
| 24     | mode_state       | uint8   | enum     | 0x00 - 0x04         |
| 25     | uptime_sec       | uint32  | s        | 0 - 2 <sup>32</sup> |

| Offset | Field           | Type   | Units       | Range            |
|--------|-----------------|--------|-------------|------------------|
| 29     | cumulative_revs | uint64 | revolutions | 0 - 2^64         |
| 37     | sequence_num    | uint32 | -           | Wrapping counter |

Table 8: Telemetry Response Packet Structure (41 bytes)

## 8. Register and Memory Map

The following registers are accessible via CONFIG\_READ (0x20) and CONFIG\_WRITE (0x21) commands. All multi-byte values are little-endian.

| Address | Field            | Type    | Access | Description                          |
|---------|------------------|---------|--------|--------------------------------------|
| 0x0000  | fw_version_major | uint8   | R      | Firmware major version               |
| 0x0001  | fw_version_minor | uint8   | R      | Firmware minor version               |
| 0x0002  | hw_revision      | uint8   | R      | Hardware revision number             |
| 0x0003  | serial_number    | uint32  | R      | Unit serial number                   |
| 0x0010  | speed_setpoint   | float32 | RW     | Current speed setpoint (RPM)         |
| 0x0014  | torque_setpoint  | float32 | RW     | Current torque setpoint (mNm)        |
| 0x0018  | mode_register    | uint8   | RW     | Operating mode (0x00-0x04)           |
| 0x0020  | watchdog_timeout | uint16  | RW     | Watchdog timeout (ms, default 500)   |
| 0x0022  | comm_timeout     | uint16  | RW     | Comm loss timeout (ms, default 2000) |
| 0x0030  | speed_kp         | float32 | RW     | Speed PID proportional gain          |
| 0x0034  | speed_ki         | float32 | RW     | Speed PID integral gain              |
| 0x0038  | speed_kd         | float32 | RW     | Speed PID derivative gain            |
| 0x0040  | torque_limit     | float32 | RW     | Max torque limit (mNm)               |
| 0x0044  | speed_limit      | float32 | RW     | Max speed limit (RPM)                |
| 0x0050  | telem_rate_hz    | uint8   | RW     | Telemetry rate (1-100 Hz)            |
| 0x0060  | fault_mask       | uint32  | RW     | Fault enable mask (1=enabled)        |
| 0x0064  | fault_register   | uint32  | R      | Current fault state (see Section 9)  |
| 0x0070  | cumulative_hours | uint32  | R      | Total operating hours                |
| 0x0074  | cumulative_revs  | uint64  | R      | Total flywheel revolutions           |
| 0x0080  | cal_offset_speed | float32 | RW     | Speed sensor calibration offset      |
| 0x0084  | cal_gain_speed   | float32 | RW     | Speed sensor calibration gain        |
| 0x0090  | inhibit_polarity | uint8   | RW     | INHIBIT pin polarity (0=active low)  |

Table 9: Configuration Register Map

## 9. Fault Register

The fault register is a 32-bit bitfield accessible at address 0x0064 and also reported in every telemetry packet. Bits are latched until cleared by a CONFIG\_WRITE to the fault\_register address or by a power cycle. Bits 16-31 form a wrapping event counter.

| Bit   | Name             | Description                                      | Recovery Action                                      |
|-------|------------------|--------------------------------------------------|------------------------------------------------------|
| 0     | OVERCURRENT      | Motor phase current exceeded 2x rated            | Auto-enters SAFE_MODE; clear fault to resume         |
| 1     | OVERVOLTAGE      | Bus voltage exceeded 38V clamp threshold         | Bus clamp active; check regen load                   |
| 2     | UNDERVOLTAGE     | Bus voltage below 7.0V minimum                   | Check power supply; auto-recovery on voltage restore |
| 3     | OVERTEMP_PCB     | PCB temperature exceeded +80C                    | Reduce duty cycle; check thermal path                |
| 4     | OVERTEMP_BEARING | Bearing temperature exceeded +75C                | Reduce speed; check thermal interface                |
| 5     | HALL_FAULT       | Hall sensor reading invalid                      | Auto-enters SAFE_MODE; may require power cycle       |
| 6     | COMM_TIMEOUT     | No valid command received within watchdog period | Auto-enters SAFE_MODE; send HEARTBEAT to resume      |
| 7     | MOTOR_STALL      | Motor speed zero despite torque command          | Check for mechanical obstruction; power cycle        |
| 8-15  | RESERVED         | Reserved for future use                          | -                                                    |
| 16-31 | EVENT_COUNTER    | 16-bit wrapping event counter                    | Increments on each fault event                       |

Table 10: Fault Register Bitfield (32-bit)

## 10. Command Examples

The following examples show complete packet bytes for common operations. All multi-byte values are little-endian. SYNC word is always 0xEB90 (bytes: 0xEB 0x90).

### 10.1 Heartbeat (Keepalive)

Request: EB 90 00 00 01 00 [CRC32]

*SYNC=0xEB90, LEN=0, MSG\_ID=0x01, SEQ=0x00, no payload, CRC32 over bytes 2-5.*

### 10.2 Set Speed to +3000 RPM

Request: EB 90 04 00 03 01 00 80 3B 45 [CRC32]

*SYNC=0xEB90, LEN=4, MSG\_ID=0x03, SEQ=0x01, Payload=3000.0 as IEEE 754 float32 (0x453B8000 LE), CRC32.*

### 10.3 Set Mode to TORQUE\_CTRL

Request: EB 90 01 00 02 02 02 [CRC32]

*SYNC=0xEB90, LEN=1, MSG\_ID=0x02, SEQ=0x02, Payload=0x02 (TORQUE\_CTRL), CRC32.*

### 10.4 Request Telemetry

Request: EB 90 00 00 10 03 [CRC32]

Response: EB 90 29 00 11 03 [41 bytes telemetry] [CRC32]

*LEN=0x29 (41), MSG\_ID=0x11, payload per Table 8.*

## 11. Notes and Constraints

1. The wheel must not be powered during launch vehicle ascent unless the pressure environment has been qualified to prevent arcing.
2. The INHIBIT\_N pin must be pulled high (or left floating with internal pull-up) for normal operation. Pulling INHIBIT\_N low disables the motor driver immediately.
3. The spacecraft power system must be capable of absorbing regenerated current during wheel deceleration. If the bus cannot absorb the energy, the 38V clamp will activate.
4. Maximum cable length between the spacecraft OBC and the wheel connector is 15 meters for RS-422 at 460,800 baud. Use shielded twisted pair cable.
5. The watchdog timer defaults to 500 ms. If no valid command (including HEARTBEAT) is received within this period, the wheel enters SAFE\_MODE automatically.
6. All firmware updates require authentication handshake before entering FW\_UPDATE mode. Contact AnyCompany for firmware update procedures.
7. The wheel should be stored in a clean environment to prevent contamination of the bearings. Storage in a sealed bag with desiccant is recommended.
8. Bearing preload is factory-set and should not be adjusted in the field. Any abnormal vibration or noise should be reported to AnyCompany.
9. The redundant RS-485 interface is active simultaneously with the primary RS-422. The wheel responds on whichever interface received the most recent valid command.
10. Telemetry rate can be adjusted from 1 Hz to 100 Hz via CONFIG\_WRITE to register 0x0050. Default is 10 Hz.